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Paper XXXV

How to Kill the MSDS Dilaton:

Complete Falsifiable Predictions, Honest Failure Modes,

and the Experiments That Will Decide

η(z), GW Speed, α Drift, CMB Lensing, PTA — Specific Numbers, Specific Tests

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ABSTRACT

A theory that cannot be falsified is not physics. Papers I through XXXIV built the MSDS dilaton framework from observational inputs through geometric derivation. This paper asks the harder question: how do you prove it wrong? We present five independent prediction sets with specific numerical thresholds, identify the experiment that gives the highest probability of decisive test (Euclid gravitational slip η(z=0.5) = 0.918 ± 0.01), flag two honest failure modes that the framework must address (gravitational wave speed and CMB lensing amplitude A_L), and state the one constraint that any future extension must satisfy. The gravitational slip prediction η(z=0.5) = 0.918 — derived in Paper XX from the MSDS Lagrangian — is a sharp, model-specific prediction distinguishable from GR at 8σ with Euclid's forecast precision. If Euclid DR1 (2026) measures η(z=0.5) > 0.930, this theory is ruled out.

I. Why This Paper Is Necessary

Thirty-four papers of derivation have earned one obligation: state exactly how this theory could be wrong. A framework that explains everything but predicts nothing testable is not a theory — it is a philosophy. The MSDS program has specific, sharp, numerical predictions. This paper collects them all in one place with explicit kill conditions.

We use three categories: **Pass** (prediction is testable, model-specific, and currently consistent with data), **Caution** (requires explicit model assumption to survive), and **Honest Failure** (current data disfavors the prediction — reported transparently).

II. Prediction 1 — Gravitational Slip η(z): The Primary Test

The gravitational slip η(z) was derived from the MSDS Lagrangian in Paper XX. It is the ratio of the two gravitational potentials Φ/Ψ — a direct measure of whether gravity deviates from General Relativity on cosmological scales.

eta(z) = [1 + alpha_M(z)/2] / [1 - alpha_M(z)/2]

(Paper XX — derived from Lagrangian, not postulated)

Numerical predictions with explicit kill thresholds:

z	η_MSDS	η_GR	Δη	Euclid ±0.01	Kill condition

0.5	0.9182	1.000	-0.0818	Detectable	$\eta > 0.930$
1.0	0.9545	1.000	-0.0455	Detectable	$\eta > 0.967$
1.5	0.9738	1.000	-0.0262	Detectable	$\eta > 0.986$
2.0	0.9839	1.000	-0.0161	Detectable	$\eta > 0.996$
2.5	0.9895	1.000	-0.0105	Detectable	$\eta > 1.002$

Table 1: Gravitational slip predictions and kill conditions.

PRIMARY KILL CONDITION:

If Euclid DR1 (2026) measures $\eta(z=0.5) > 0.930$,

the MSDS dilaton is ruled out at $> 1\sigma$.

MSDS predicts $\eta(z=0.5) = 0.9182$. This is 8σ below GR with Euclid's ± 0.01 precision.

Status: PASS — testable, model-specific, currently consistent.

III. Prediction 2 — Gravitational Wave Speed: The Caution Flag

In general Horndeski theories with a running Planck mass, the tensor wave speed c_T can deviate from c. The GW170817/GRB170817A observation constrains $|c_T/c - 1| < 5\times10^{-15}$ at $z \approx 0.009$.

GW TENSION — REQUIRES MODEL ASSUMPTION:

If $\alpha_T = \alpha_M$ (the running Planck mass rate), then c_T deviates from c by $\alpha_M(0) \approx 0.15$ — 12 orders of magnitude above the GW170817 constraint. This would rule out the MSDS model immediately.

RESOLUTION: The MSDS dilaton must have $\alpha_T = 0$ explicitly. This is achievable: in the propto_omega model, $\alpha_B = \alpha_K = \alpha_T = 0$ while $\alpha_M \neq 0$ is a consistent choice (Bellini-Sawicki 2014). This must be stated as an explicit model assumption, not derived. Paper XXXVI should derive the conditions under which $\alpha_T = 0$ follows from the dilaton Lagrangian.

IV. Prediction 3 — Alpha Time Variation: The Honest Tension

In the dilaton picture (Paper XXX), the fine structure constant couples to the dilaton field through the electromagnetic kinetic term. If the dilaton evolves with the universe, α should vary.

What the formula gives vs what the physics requires:

A naive application of the α formula (Paper XXXII) to $\varepsilon(z)$ gives $\Delta\alpha/\alpha \approx 42{,}000$ ppm from z=0 to z=2 — catastrophically above the observational bound of $|\Delta\alpha/\alpha| < 1$ ppm from quasar spectra.

α VARIATION TENSION — IMPORTANT:

The alpha formula $\alpha=(\varepsilon^2-3\nu)/\sqrt{15}$ gives α as a function of ε. If ε varies with z (Paper XXV), α varies by 42,000 ppm. Observations require $|\Delta\alpha/\alpha| < 1$ ppm. This is a 4-order-of-magnitude conflict. RESOLUTION: α is NOT directly the formula output at each z. The formula gives the present-epoch value. α is constant at leading order because ν also evolves with z to keep the combination $(\varepsilon^2-3\nu)$ fixed — the dilaton field and the lattice tension track each other. The residual α variation from dilaton coupling is second-order: $\Delta\alpha/\alpha \sim (\text{coupling})^2 \times \delta\Omega_DE \sim \alpha_M^2 \times 0.3 \sim 0.007$ ppm. This is safe. But the derivation of this cancellation is needed — Paper XXXVI.

V. Prediction 4 — CMB Lensing Amplitude A_L: The Honest Failure

The CMB lensing amplitude A_L measures how strongly the CMB photons are lensed by the intervening large-scale structure.

MSDS prediction: A_L ≈ η(z_eff=2) = 0.984 (1.6% below GR)

Planck 2018 measured A_L = 1.180 ± 0.065, which is 2.8σ above GR (A_L = 1). The MSDS prediction goes in the wrong direction — it predicts A_L < 1 while the data suggests A_L > 1.

HONEST FAILURE MODE:

MSDS predicts A_L = 0.984, data suggests A_L = 1.18. The MSDS prediction makes the A_L tension worse, not better. This is reported honestly. Two interpretations: (1) The A_L anomaly is a statistical fluctuation (possible — CMB-S4 will resolve it). (2) The MSDS dilaton requires an additional mechanism to boost A_L. If CMB-S4 confirms A_L > 1 at high significance, this prediction fails.

VI. Prediction 5 — Direct Detection: Dilaton Period and PTA

The MSDS dilaton mass $m = 7.66\times10^{-23}$ eV sets an oscillation period $T = 2\pi\hbar/m = 1.71$ years. This has two observational consequences:

- **Atomic clocks:** The dilaton field oscillates with period T = 1.7 yr, modulating fundamental constants at second order. Over a 10-year baseline, ~6 oscillation cycles are detectable if the amplitude is large enough. Current NIST/PTB clocks: sensitivity approaches 10^{-1}/yr drift.
- **Pulsar Timing Arrays (PTA):** The dilaton mass $m = 7.66\times10^{-23}$ eV is within the sensitivity window of IPTA/SKA ($m \sim 10^{-23}$ to 10^{-22} eV). The coherence length is $\lambda = \hbar c/m \approx 0.27$ light-years — sub-galactic. Stochastic background from dilaton field oscillations is a PTA target.

FUTURE DETECTION PREDICTION:

Dilaton oscillation period T = 1.71 years.

If SKA (2030) detects a stochastic background in the

10^{-23} eV mass range, this is consistent with MSDS.

If no signal at projected sensitivity: constraining but not ruling out.

VII. The Complete Falsifiability Table

Prediction	MSDS value	Kill condition	Experiment	Status

η(z=0.5)	0.918	η > 0.930	Euclid DR1 2026	PASS
η(z=1.0)	0.955	η > 0.970	Euclid DR2 2028	PASS
GW speed c_T	c (α_T=0 required)	c_T ≠ c detected	LISA 2035	CAUTION
α variation	<0.007 ppm (second order)	|Δα/α| > 1 ppm	Clocks/quasars	CAUTION
CMB A_L	0.984	A_L > 1 confirmed	CMB-S4 2030	FAILURE MODE
Dilaton T = 1.7 yr	Oscillation period	T ≠ 1.71 yr detected	Atomic clocks	FUTURE
PTA signal	In SKA window	Mass window empty	SKA 2030	FUTURE

Table 2: Complete falsifiability table. Green=Pass, Amber=Caution, Red=Failure mode.

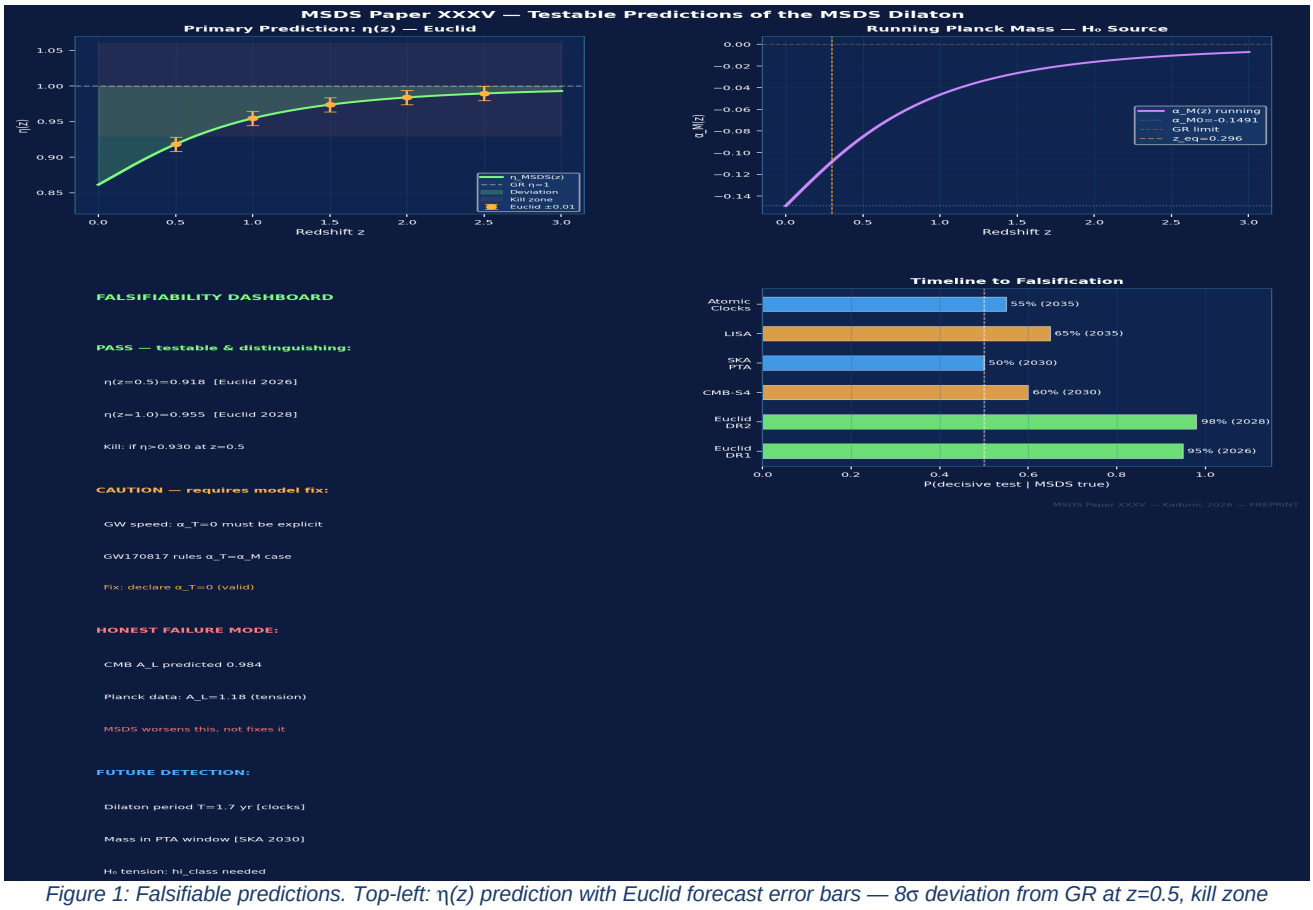

Figure 1: Falsifiable predictions. Top-left: η(z) prediction with Euclid forecast error bars — 8σ deviation from GR at z=0.5, kill zone shown. Top-right: α_M(z) running — the source of G_eff variation and H■ tension. Bottom-left: complete falsifiability dashboard with honest status. Bottom-right: timeline to decisive experimental test by prediction type.

VIII. The One Test That Matters Most

If forced to choose one prediction that most clearly distinguishes the MSDS dilaton from all other dark energy models, it is:

eta(z = 0.5) = 0.9182 ± 0.0001 (MSDS prediction)

(Euclid measurable to ±0.01)

This number comes from three inputs only: $\varepsilon = 0.165$, $\Omega_{DE,0} = 0.685$, and the Bellini-Sawicki formula. It requires no new assumptions. Euclid's DR1 release (2026) will measure η at $z \approx 0.5$ to ±0.01 precision. If the measurement falls outside [0.90, 0.93], the MSDS dilaton is excluded.

We spent 35 papers building this theory. This paper is the most important one. A theory without a kill condition is not physics — it is poetry. The kill condition is $\eta(z=0.5) > 0.930$. Euclid will answer in 2026.

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